



Association of Zoos & Aquariums

2020 AZA CONSERVATION GRANTS FUND

Application Instructions

DEADLINE

This 2020 Conservation Grants Fund (CGF) electronic application must be used for all proposals. Completed applications must be submitted via email no later than **5:00 p.m. EDT** on **16 March 2020**. An auto-response email with a time-stamp will confirm receipt of the application.

*Incomplete applications, or applications received after 5:00 p.m. EDT on 16 March 2020, **regardless of cause**, will NOT be eligible for consideration. NO DEADLINE EXCEPTIONS WILL BE MADE.*

MEMBERSHIP

The Principal Investigator (PI) and all Co-Investigators (CIs) *must be individual AZA members* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership and individual members at the Friend level are not eligible. Other participants actively involved in the completion of the project should be listed as collaborators and are not required to be AZA members. To apply for membership, visit <https://www.aza.org/individual-membership>.

FUNDING TIMELINE

All CGF projects must be completed within 12 months. CGF grants are intended to serve as seed money to attract additional funding and proposals should request funds only for this one-year period. Identical components of a project will not be funded for more than two grant cycles; however, funding requests for different components of the same project may be applied for in subsequent grant cycles.

CGF proposals undergo a two-tiered review process, outlined at <https://www.aza.org/cgf-selection-process>. Applicants will be notified of funding decisions in early September 2020 and funds will be available 1 October 2020. CGF funds can only be used for proposal expenses incurred on or after the award date. Expenses incurred prior to the award date will not be covered by CGF funds.

APPLICATION GUIDELINES

The CGF application and all supplemental forms are posted online at: <http://www.aza.org/cgf-information-for-applicants/>. These documents contain form fields that must be completed and submitted via email to: CGFapp@aza.org. *All application materials must be in electronic format. No hard copies of any application materials will be accepted.*

It is strongly recommended that you save a copy of your completed application and supplemental forms as a backup. *Only* completed applications, submitted by the deadline defined above, will be reviewed. Applicants will be informed electronically if their application is incomplete.

APPLICATION INSTRUCTIONS AND CHECKLIST: All completed application materials must be submitted **in one email** by the deadline identified above and include the following:

CGF Application: The application is one document consisting of the three sections outlined below. All sections must be complete in order for the application to be considered complete. The application must be submitted as a Microsoft Word document titled “**Application**” and attached to your CGF Application email.

Applications submitted in Portable Document Format (PDF) will NOT be eligible for consideration. NO EXCEPTIONS WILL BE MADE.

Note that “spell checker” and “track changes” functions cannot be used in this form. AZA strongly recommends that applicants compose the content for their Project Abstract and Project Proposal in a separate document to make use of these features, and then copy and paste the final text into the form fields. When composing in a separate document, be aware of character limitations. If the copied text exceeds the character limit when pasted into a form field, *all text exceeding the character limitation will be cut off and will not appear in the final proposal.*

Also note that certain formatting such as italics, bold, and underlining, is unavailable due to the application format. AZA recommends the use of numbering and spacing to emphasize important points in the text.

- ☒ **Cover Sheet.** Fill in the text fields for the project title, amount requested, and PI and CI information. Choose *two* project categories from the given list. Include a project abstract no longer than 1,500 characters, including spaces.
- ☒ **Project Proposal.** Text fields are provided for responses to questions 1-12. Fields have a limit on the number of characters that may be used to keep the application concise. Character limits, including spaces, are stated following each question. If a question is not applicable to your proposal, write “N/A” in the field but *do not leave any form fields blank*, as this may delay processing of your application.
- ☒ **Budget.** Required budgetary information includes:
 - ☒ Detailed Project Budget. Detail how the budget will be allocated to specific items and indicate which funding source will pay for each item. State the total project budget, the amount requested from CGF, and other financial support applied for or received. Each of these amounts should reflect the one-year duration of the project. Funding decisions are announced in early September and CGF funds are to be used for work from the announcement date forward. Expenses incurred prior to the announcement will not be covered by CGF funds.
 - ☒ Budget Justification. Provide an explanation of why specific major items are essential to the project and reasonable in cost.

Required Attachments: The following must be attached to your CGF Application email:

- ☒ **Curricula Vitae.** A curriculum vitae (CV) is required for **each** PI and CI listed on the Cover Sheet and must follow the format stated in these guidelines. CVs must not exceed a **maximum** length of two letter-sized (8.5 by 11 inch) pages each and must be typed in a legible font size (12 point recommended); each side of the paper counts as one page. CVs may be attached for collaborators but are not required. Each CV must be submitted as a separate document entitled, “**CV Investigator’s Last Name.**” *CVs that exceed the maximum length will be cut off at the second page.*
- ☒ **Statement of Institutional Support Forms.** Electronically signed *Statements of Institutional Support* from the CEO(s)/Institutional Director(s) (not the COO, General Curator, Vice President, Dean, or Department Director) of the PI’s institution and **EACH** CI institution *and* collaborating institution must be attached to the CGF Application email as separate documents titled “**SIS**

Institution's Name. The *Statement of Institutional Support* form can be found with the rest of the CGF Application materials on the AZA website identified above.

Applicants wishing to include support letters from entities *other than* PI and CI institutions, collaborating institutions, or AZA Animal Programs may submit such items as appendices.

Note that letters of support from PI and CI institutions and collaborating institutions will not be accepted in lieu of the *Statement of Institutional Support* form. In addition, letters of support/endorsement **from AZA Animal (Taxon Advisory Groups, Species Survival Plans®, etc.) and SAFE Programs** are not accepted although coordination with AZA Animal and SAFE Programs is encouraged and should be described in Question 1. To assure Animal and SAFE Program review of all relevant CGF submissions, AZA Program Leaders will serve as expert reviewers in the CGF application review process and may express their support for a proposal at that time.

- ☒ **Permits.** If permits are required to complete your proposed project, you must attach a copy of each to your CGF Application email. Each permit attached to the CGF Application email must be titled **"Permit #."** *If you are not able to obtain a required permit before submitting this proposal, you must submit it by 15 June 2020 to ensure award eligibility.*
- ☒ **Proposal Resubmission Form.** If a proposal for this project was previously submitted for CGF funding and rejected, the revised proposal *must* include a completed *Proposal Resubmission Form* as an attachment on the CGF Application email. This attachment must be submitted as a Microsoft Word document titled **"Resubmission Form"** and attached to your CGF Application email. *Resubmission forms submitted in Portable Document Format (PDF) will not be accepted.* The form can be found with the rest of the CGF Application materials on the AZA website identified above.

Note: Proposals that fail to provide all items identified in the above checklist in the correct format will not be reviewed.

Non-Required Attachments: The following attachments to your CGF Application email are optional:

- ☒ **Citations (recommended).** Citations/references for statements made in the Project Proposal should be included in a separate document titled **"Citations"** and attached to the CGF Application email.
- ☐ **Appendices.** Appendices are allowed, but may only *clarify*, and not *supplement*, information in the narrative. For example, photos or diagrams would be acceptable, but not videos or publications. Appendices must be attached to the CGF Application email titled **"Appendix #."**

Note: Appendices that do not adhere to these guidelines will be removed from the proposal prior to review.

SUBMISSION INSTRUCTIONS

By submitting your application, you agree, if funded, to abide by the Grant Terms and Conditions posted on AZA's website at <http://www.aza.org/cgf-information-for-applicants/>.

Proposals must be submitted electronically using the 2020 CGF Application and related forms via a **single email, with the proposal title as the subject line**, to CGFapp@aza.org no later than 5:00 p.m. EDT on 16 March 2020. Hard copies of any application materials or applications received after this deadline will not be accepted *regardless of cause*.

AZA's email system accepts messages up to 8 MB in size. In cases when applications are too large to be accepted in one email, AZA will accept the application submission in two separate emails, provided that the same proposal title is identified in the subject line of each.

The 2020 CGF Application email **must** include the following attachments

- CGF Application (this completed form);
- CVs for each Principal and Co- Investigator(s);
- Statements of Institutional Support for each PI and CI institution and collaborating institution; and
- A Proposal Resubmission form, if applicable.
- The Application email may also include citation and appendices attachments.

Auto-response emails will be sent to applicants confirming receipt of the application package. If an auto-response is not received, please resubmit the application. If an applicant submits more than one proposal, s/he should ensure that an auto-response was received for *each* submission. If submission problems arise, contact the AZA office at 301-562-0777. If you submit your application after the 5:00 p.m. deadline, you may still receive an auto-response from the CGF Application inbox; however, your proposal will not be reviewed.

Applications which fail to follow submission instructions or are time-stamped after the deadline will not be considered and applicants will be notified that their application is inadmissible.

APPLICATION ASSISTANCE

The CGF website (<http://www.aza.org/cgf-information-for-applicants/>) contains a wealth of information to help applicants compose a competitive proposal.

Proposals received before 7 March will be reviewed by AZA staff to ensure that the application is completed correctly. Early submission allows time for AZA staff to inform applicants if problems are noted in time for resubmission. Please note that AZA staff does not review early proposals for content.

2020 AZA CONSERVATION GRANTS FUND Cover Sheet

Project Number: 20-1675
For AZA use only

Title of Project: Improved Hawaiian monk seal conservation through genomics

Amount Requested: \$18,345 (round to the nearest whole dollar)

Project Start Date: 10/1/20

End Date: 9/30/21

(CGF funding is for one year only, beginning no earlier than the award date)

Principal Investigators: Each Principal Investigator (PI) and Co-Investigators (CI) *must be an individual AZA member at the time of application* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership. Other participants should be listed as collaborators on this form.

A Curriculum Vitae (CV) is required for each PI and CI, and a Statement of Institutional Support (SIS) form is required for each PI and CI Organization.

1. Principal Investigator (PI) Name and Title: Dr. Emily Latch, Professor

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (required; available at:

<http://www.aza.org/cgf-information-for-applicants/> ☒ (required)

Individual AZA Member Number: 81026762 (required)

Institution/Organization: University of Wisconsin, Milwaukee

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: 3209 N. Maryland Ave, Lapham Hall, Rm S181, Milwaukee, WI 53211

Phone: (414) 229-4245

E-mail: latch@uwm.edu

2. Co-Investigator (CI) Name and Title: Dr. Samantha Hauser, Postdoctoral Fellow

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (available at:

<http://www.aza.org/cgf-information-for-applicants/> ☒ (required)

Individual AZA Member Number: 81054100 (required)

Institution/Organization: University of Wisconsin, Milwaukee

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: 3209 N. Maryland Ave, Lapham Hall, Rm S181, Milwaukee, WI 53211

Phone:

E-mail: hausers@uwm.edu

3. Co-Investigator (CI) Name and Title:

CV Attached? ☐ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (available at:

<http://www.aza.org/cgf-information-for-applicants/> ☐ (required)

Individual AZA Member Number: (required)

Institution/Organization:

Statement of Institutional Support Form Attached? ☐ (required)

Institutional Address:

Phone:

E-mail:

Note: If additional Co-Investigators need to be included, add their information in a separate document titled “**Additional Co-Investigators**” and include it as an attachment to your CGF Application email.

Collaborating Institutions or Organizations: A collaborating institution is defined as any institution actively involved in the design or completion of the project. Funding institutions *not* actively involved in project design or completion should be listed in the “Grant/Matching Support” section of the Project Budget and need not be listed as collaborators.

List all collaborating institutions below, including the PI’s and CI’s institutions listed on the Cover Sheet, as well as those associated with the personnel listed in Question 6, and attach a Statement of Institutional Support form for each collaborating institution to the CGF Application email.

Collaborating Institution Names:

University of Wisconsin, Milwaukee *Statement of Institutional Support Form Attached?* ☒ (required)

National Oceanic and Atmospheric Administration Fisheries Pacific Islands Fisheries Science Center - Protected Species Division *Statement of Institutional Support Form Attached?* ☒ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Note: If additional Collaborating Institutions or Organizations need to be included, add their information in a separate document titled “**Additional Collaborators**” and include it as an attachment to your CGF Application email.

Project Categories (choose two that best describe the project):

☐ Animal/Population Management (*Ex situ*)

☒ Research

☐ Conservation Education

☐ Animal Health

☒ Field Conservation (*In situ*)

☐ Animal Welfare

Project Abstract (limit 1,500 characters with spaces):

The Hawaiian monk seal (*Neomonachus schauinslandi*) is a critically endangered species with small remnants scattered across the Hawaiian archipelago. As such, there is an immediate need to enhance in-situ conservation efforts and explore captive population management. The goal of the proposed research is to survey existing genetic variation in the Hawaiian monk seal across the Hawaiian archipelago. This project serves two critical and complementary purposes: 1) to immediately improve ongoing recovery efforts in the wild, and 2) to design the establishment of a Hawaiian monk seal captive breeding program. We will use a powerful genomic approach (i.e., a species-specific SNP panel) to estimate genetic diversity, delineate populations, and characterize movement among populations. This survey of genetic variation in wild Hawaiian monk seals will allow us to identify populations at highest risk of extinction or that warrant targeted conservation efforts. We will also leverage these genetic data to design the establishment of a captive breeding program by determining the optimal number and representation of founders. Although genetic considerations are only one component of a successful breeding program, genetic composition of founders is important to ensure long-term retention of genetic variation. With our partners at NOAA-NMFS, we have a unique opportunity to integrate in-situ and ex-situ conservation to best support Hawaiian monk seal conservation.

2020 AZA CONSERVATION GRANTS FUND

Project Proposal

1. Describe the overall project and its significance for conservation and/or management. What species, sites, habitats, populations, or questions are targeted or addressed and what critical needs will be met? Explain the project's relevance for AZA animal/conservation/education programs or institutions, and/or how it reflects the priorities of the AZA Board (e.g., sustainability of managed populations, SAFE: Savings Animals From Extinction) or of related Animal or SAFE species programs. **LIMIT 5,000 CHARACTERS with spaces**

The Hawaiian monk seal is one of the world's most endangered marine mammals, with approximately 1,200 individuals left globally. The Hawaiian monk seal has faced two major population crashes due to historical hunting (Rice 1960) and continued declines from habitat loss, marine debris, shark predation, resource competition from commercial fisheries, and male aggression toward females and pups (Antonelis et al. 2006). These declines have left the Hawaiian monk seal in small populations scattered across the Hawaiian archipelago. Small, isolated populations are not only less demographically stable than large populations, but they are also more susceptible to erosion of genetic variation. A lack of genetic variation not only makes a population more susceptible to inbreeding depression (Ralls et al. 1979, Cmokrak and Roff 1999, Keller and Waller 2002), but also less able to adapt to changing environmental conditions (Falconer 1981, Keller et al. 1994, Willi et al. 2006, Markert et al. 2010). Preserving genetic variation has become a priority for management, particularly for small and isolated populations, in order to maintain long-term viability (McNeely et al. 1990, Lacy 1997; NOAA-NMFS, 2007).

For more than two decades, intensive conservation efforts have helped the Hawaiian monk seal avoid extinction. Under the purview of the National Oceanographic and Atmospheric Administration's National Marine Fisheries Service (NOAA-NMFS), the Hawaiian Monk Seal Research Program works to enhance the recovery of Hawaiian monk seals through research, monitoring, and mitigating factors impeding species recovery. Ongoing conservation efforts include protection of marine areas (e.g., to mitigate contamination, human disturbance, shark predation, fishery interactions), removal of aggressive males (those injuring or killing females and pups), ongoing species monitoring, vaccination programs, and rehabilitation of injured and diseased individuals at the Marine Mammal Center (NOAA-NMFS, 2007). It is likely that a captive breeding program will need to be established to ensure long-term persistence of the Hawaiian monk seal, and the Research Program and Hawaiian monk seal conservation partners are interested in exploring options for genetic management in captivity (Appendix I).

In partnership with NOAA-NMFS, the goal of this project is to survey the genetic diversity in Hawaiian monk seals across the Hawaiian archipelago. These data serve two critical and complementary purposes relevant to species survival: 1) to immediately improve ongoing recovery efforts in the wild, and 2) to design the genetic establishment and management of a Hawaiian monk seal captive breeding program. We are in a unique position to simultaneously integrate in-situ and ex-situ conservation to best support Hawaiian monk seal conservation (Ogden et al. 2017). In the wild, genetic data will help us to identify genetically unique populations, guide translocations and head start programs, and understand patterns of movement among islands. For example, these genetic data could be used for genetic rescue of a population with high inbreeding, or to facilitate preservation of genetic variation through increased genetic exchange (natural or human-mediated) between populations. We will leverage the same genetic data to design a captive breeding program that determines the optimal number and representation of founders to adequately sample genetic diversity from the wild and to retain the most genetic variation over time. Genetic considerations are just one part of a successful captive breeding program including husbandry, reproductive physiology, and behavior, but nevertheless are an important step toward successful establishment of a captive breeding program. The Hawaiian monk seal would benefit from a carefully planned captive breeding program in advance of need, as would other species considering captive population sustainability initiatives.

The proposed research is compatible with AZA-CGF priorities through its collaborative effort, particularly with the NOAA-NMFS Hawaiian Monk Seal Research Program, to address essential and immediate conservation concerns for a critically endangered species. A central mission of AZA is to use molecular data for in-situ conservation of small populations, which is at the core of our project. Lastly, AZA strives to support projects that provide basic scientific

research while simultaneously improving in-situ and ex-situ conservation. The data we collect on wild populations will advance in-situ conservation, and provide critical information for establishing a captive breeding program that captures the full range of genetic variation remaining in Hawaiian monk seals. While rare for many endangered species, rangewide genetic data is the gold standard for population establishment because it maximizes our ability to retain genetic variation and our potential to save species from extinction over the long term.

2. List the goal(s) and associated objective(s) of the proposed project and the actions that will be taken to achieve them. Goals should describe the desired status in the long-term of the question or problem that project will address. Objectives should state the immediate results that are necessary to meet the long-term goal(s). Actions should be the specific activities the project will undertake to complete each objective. If the proposal is a request to fund part of a larger project, this section should focus on the part for which CGF funding is being requested during the one-year duration, while also providing some context regarding the larger effort. **LIMIT 4,500 CHARACTERS with spaces**

Broadly, our goal is to improve Hawaiian monk seal recovery by surveying genetic variation across the Hawaiian archipelago. These data serve two critical and complementary purposes relevant to species survival: 1) to immediately improve ongoing recovery efforts in the wild, and 2) to design the establishment of a Hawaiian monk seal captive breeding program. To meet our goal, we will generate thousands of single nucleotide polymorphisms (SNPs), individual DNA base variants located across the genome. SNPs are powerful for conservation and management; they deliver an unprecedented wealth of data spanning the entire genome of an animal, providing more accurate estimates of key population parameters and insight into complex population history. SNPs are a particularly good tool for endangered species with low genetic diversity like the Hawaiian monk seal (Shultz et al. 2009), and the recent availability of a Hawaiian monk seal reference genome makes this an exciting time to tackle some of the conservation challenges for this species. We are already in the process of developing a panel of SNPs for Hawaiian monk seals as part of a complementary project to quantify variance in reproductive success in local populations. This panel is being designed with the larger-scale survey proposed here in mind, and will be completed prior to the start of this project.

Objective 1: Survey genetic diversity in Hawaiian monk seal populations across the Hawaiian archipelago and characterize movement (gene flow) among populations.

Using the SNP panel we are currently developing as part of a complementary project, we will genotype approximately 200 Hawaiian monk seals (~20/island). We will estimate genetic diversity metrics for each island, assess inbreeding, and quantify changes in population size (N_e); SNPs have high power to detect even subtle changes to N_e (Schultz et al. 2016). We will estimate the level of genetic differentiation between islands, to better understand how Hawaiian monk seals move among islands.

The genetic diversity data we collect will provide immediate results including: a) a benchmark for future conservation of genetic diversity, b) list of at-risk populations with current inbreeding and/or low genetic diversity. The data we collect on genetic differentiation among islands will similarly provide immediate results including: a) identifying islands with genetically unique seal populations, and b) quantifying the amount and pattern of migration (gene flow) among islands. Taken together, these data will provide information about the current status of Hawaiian monk seal populations across the entire species range, and immediately improve in-situ conservation by helping to direct conservation efforts to populations that would benefit most.

Objective 2: Design the establishment of a Hawaiian monk seal captive breeding program.

Captive breeding is a useful tool to aid species recovery. Often used as a last resort, captive breeding programs can fail without sufficient planning in advance of need. We propose to use genetic data to design the establishment of a Hawaiian monk seal captive breeding program to represent the full range of genetic diversity found within the species. We will determine the optimal number of founders to adequately sample diversity in the species (e.g., 90% of current gene diversity in the total population). Our SNP data will reveal any spatial genetic variation that should be represented in the captive breeding program. Using the software PMx, we will model different proposed scenarios, including different founder numbers and diversity. We will further simulate captive breeding program outcomes for 100 years, with the goal of maintaining e.g., 90% gene diversity through prioritized pairings that minimize mean kinship. Mean kinship based strategies are highly effective at retaining variation in small and isolated populations, both in captivity and in the wild. Immediate results include a plan for a Hawaiian monk seal captive breeding program that maximizes the representation of genetic variation from wild populations during initial establishment and retains the most genetic variation in the captive population over 100 years. This approach can be readily adapted to other species seeking to establish captive breeding programs, and we aim to provide resources to facilitate application of our approach to other species..

3. Explain in detail the methodology for this project and why it is appropriate. If this is a research project, clearly state research question(s) and how it will be tested. For applied conservation projects, explain how project actions will be undertaken and proposed actions are appropriate for addressing the conservation issue addressed. If this is a conservation education or professional training project, describe the methods as they relate to the target audience, number of people that will be reached, etc. **LIMIT 7,000**

CHARACTERS with spaces

The overall goal of this project is to improve recovery of the critically endangered Hawaiian monk seal. Conservation efforts have reduced the impact of several of the most direct threats, and yet the population continues to decline and captive breeding must be considered. We have a unique opportunity to survey genetic variation in wild Hawaiian monk seals, toward two objectives: First, to identify populations that might be at highest risk for extinction (i.e., low genetic diversity), or warrant targeted conservation activities (i.e., harbor unique genetic variation). Second, we can use genetic diversity data from wild populations to plan the establishment of a captive breeding program, to ensure that we effectively capture the species' genetic diversity. Below we provide detailed methodology, including names of specific products, reagents, and software protocols where relevant. Although this increases the amount of jargon in this section, we felt it necessary to be specific about our approach.

Sample Collection: Tissue samples from islands across the Hawaiian archipelago have already been collected by the NOAA-NMFS Hawaiian Monk Seal Research Program. We will use samples from approximately 200 seals (~20 from each island). As part of our complementary project, we have intensively sampled populations of Hawaiian monk seals on Hawaii and Kauai (n=135), and we anticipate that we will be able to use a subset of these samples to represent the genetic variation on those islands.

Genotyping: We will analyze the samples using the same methods that we developed for our ongoing complementary study. We will extract DNA from tissue samples using the Qiagen DNeasy Blood and Tissue Kit, and quantify it using a Qubit fluorometer. The DNA from all samples will be normalized to the same concentration; this maximizes the overall quality of the resulting data by ensuring that when pooled together, no one sample is over-represented. We will genotype individuals using the BestRAD (Ali et al. 2016) genotyping-by-sequencing protocol that yields thousands of SNP markers. In this approach, samples are cut with a restriction enzyme into DNA fragments, a known sequence is attached onto the fragments (both to increase product and to identify individual seals), fragments are further sheared to a standardized size range (250-500bp) and pooled together, and then are sequenced using next-generation sequencing (e.g., Illumina HiSeq). This process generates millions of DNA sequences from which we can genotype each individual for subsequent data analyses. In our complementary project, we are currently developing a panel of highly informative genomic markers for parentage assignment, genetic diversity estimation, and population structure characterization in Hawaiian monk seals. We have optimized laboratory protocols in our lab, and developed a bioinformatic pipeline to assemble and align the massive amounts of sequencing data generated, identify the SNPs, and genotype individual animals. The existence of a Hawaiian monk seal genome enhances the quality of our project by allowing for the discovery of more SNPs than without a genome, resulting in higher analytical power and better resolution. We will analyze the suite of informative SNP markers for both projects to maximize the overlap and impact of our results.

Data Analysis:

Objective 1: A suite of several thousand genome-wide SNPs is expected to outperform other types of genetic markers (e.g., microsatellites) in terms of accuracy in quantifying patterns of genetic diversity and genetic structure, especially in populations with low genetic variation like the Hawaiian monk seal (Morin et al. 2009, Nazareno et al. 2017). We will estimate genetic diversity metrics including allelic richness, heterozygosity, and inbreeding for each population of Hawaiian monk seals using the software R. We will also estimate the effective population size (N_e) of each population, a critical parameter used in population recovery.

We will estimate genetic differentiation among Hawaiian monk seal populations using complementary metrics of SNP differentiation (D) and fixation (G_{st}). These metrics will paint a picture of how Hawaiian monk seals from different islands are related to one another and how often (or how recently) they shared genes. We will further explore genetic differentiation among islands by identifying any groups of islands that are genetically similar using the software STRUCTURE. We will quantify the amount and pattern of movement (gene flow) among islands using the complementary software programs BayesAss, which estimates gene flow, and GeneClass, which identifies recent migrants. We will create a connectivity network to visualize seal movement among islands. Overall, these analyses will tell us whether, and how often, Hawaiian monk seals are moving between islands. Our hypothesis is that seals move between adjacent islands more frequently than between distant islands, but geographic distance may not be the only factor influencing gene flow among islands.

Objective 2: We will use SNP data collected for Objective 1 to design a strategy for effectively establishing a captive breeding program. In Hawaiian monk seals, we have a unique opportunity to properly design the establishment of a captive breeding program before it is an emergency. We will determine the optimal number and representation of founders to adequately sample genetic diversity (e.g., sample 90% of the contemporary gene diversity in the species) by modeling alternative establishment scenarios, including different numbers of founders and founder diversity in the software simuPOP. We will also simulate retention of genetic variation within a captive breeding program over 100 years, with a goal of maintaining a certain level of genetic variation (e.g., 90% of gene diversity, the current goal of most SSP programs) through prioritized pairings that minimize mean kinship in the software PMx. The recommended scenario will be the one that maximizes the representation of genetic variation from wild populations during initial establishment and retains the most genetic variation in the captive population over 100 years.

We anticipate that our approach for Hawaiian monk seals, using genetic data to guide the establishment of a captive breeding program, will be broadly applicable to other endangered species that might require captive breeding programs for long-term survival. To that end, we aim to make our resources – laboratory protocols, bioinformatic pipelines, code we develop – freely available to the AZA community and conservation groups, including those focused on utilizing molecular data for conservation - AZA's Molecular Data for Population Management Scientific Advisory Group, The Wildlife Society's Molecular Ecology Working Group, the Society for Conservation Biology's Conservation Genetics Working Group, and IUCN's Conservation Genetics Specialist Group.

4. If previous CGF funding was obtained for this or a related project, summarize relevant findings and identify how they relate to the current proposal. **LIMIT 1,000 CHARACTERS with spaces**

Not applicable

5. Describe the timeline in which components of the project will be completed within the one-year project duration. Use of bulleted/numbered lists is encouraged. **LIMIT 1,500 CHARACTERS with spaces**

2019

October:

- Samples received at University of Wisconsin – Milwaukee from collaborator Dr. Stacie Robinson

November - December:

- DNA extraction and quantification for all samples
- Normalization of DNA samples to equivalent concentrations for optimal downstream performance
- Quarterly report provided to collaborators

2020

January- February:

- BestRAD laboratory work, specifically library preparation
- Quarterly report provided to collaborators

March:

- BestRAD data processing

April -June:

- Genetic diversity analysis, including allelic richness, heterozygosity, inbreeding, and Ne
- Population analysis (genetic differentiation among islands)
- Connectivity analysis, including estimating gene flow among islands and creating a connectivity network
- Quarterly report provided to collaborators

July-August:

- Captive breeding simulations to determine optimal number and representation of founders
- Simulations to determine proportion of genetic diversity retained over 100 years
- Presentation given to Hawaiian Monk Seal Research Program
- Dissemination of Hawaiian monk seal results to collaborators and conservation groups

September:

- Final report provided to collaborators
- Dissemination of resources for establishing captive breeding program that effectively samples genetic diversity from the wild, for use in other species

-Submission of manuscript to peer-reviewed journal

6. Define the project team by identifying the personnel included for this project, their roles, qualifications, and the key skills they bring. Discuss the PI and each CI listed on the Cover Sheet and any collaborators (persons actively involved in the completion of the project). Each collaborator's institution should be listed under **Collaborating Institutions or Organizations** on the application Cover Sheet. **LIMIT 2,500 CHARACTERS with spaces**

Dr. Emily Latch, PhD (Principal Investigator) is an Associate Professor at the University of Wisconsin-Milwaukee. Her lab's research investigates the evolutionary responses of populations to intensive management (in wild, semi-wild, and captive zoo populations) by studying the genetic effects of alternative management strategies, evaluating reintroduction schemes and post-release monitoring, and developing better genetic management approaches for captive populations. She has extensive experience using large genomic datasets, and has developed conservation and management plans for long-term sustainability of genetic diversity. Dr. Latch will oversee the proposed research and provide expertise regarding genomic analyses and captive conservation.

Dr. Samantha Hauser, PhD (Co-Principal Investigator) is a postdoctoral research associate in Dr. Emily Latch's research group. She has expertise in wildlife population genomics, particularly with endangered species. Her research has integrated elucidating population structure and connectivity with conservation programs. Dr. Hauser will be responsible for performing laboratory work and data analyses for the project. Dr. Hauser is currently performing the laboratory and analytical work to develop the Hawaiian monk seal SNP panel for the complementary project we have described in this proposal.

Dr. Stacie Robinson, PhD (Collaborator) is a Research Ecologist for the Hawaiian Monk Seal Research Program at the National Oceanic and Atmospheric Administration's National Marine Fisheries Service (NOAA-NMFS). She has coordinated and performed Hawaiian monk seal research and management for 5 years. Dr. Robinson will be providing expertise on Hawaiian monk seal biology, population trends, current and historical recovery efforts, and will provide the tissue samples used in our genomic research.

7. Describe the project's monitoring and evaluation plans. Explain how the goals and objectives of the project will be met, which indicators will be used, and how those indicators will be measured. If this project has an educational or professional training component, the educational impact should be discussed (ideally including impacts on conservation-related knowledge, attitudes/affect, and behavior). Plans may be quantitative or qualitative, and should differentiate between monitoring (ensuring proposed actions are taken) and evaluation (whether the project had the desired impact on the issue or question addressed). **LIMIT 3,000 CHARACTERS with spaces**

Drs. Emily Latch and Samantha Hauser will communicate with collaborator Dr. Stacie Robinson on a weekly basis during the sampling design and distribution stage. Once DNA and data analyses are underway, Drs. Latch and Hauser will communicate with Dr. Robinson monthly to ensure that project milestones, as per the timetable, are met. Additionally, Drs. Latch and Hauser will communicate on a weekly basis throughout project to review and adjust analyses as necessary. Their proximity (in the same building at the same institution) facilitates frequent interaction. Co-PI Dr. Hauser will carry out the majority of the project, so Drs. Latch and Robinson will monitor project completion and ensure that milestones are met, as per the timetable, including: 1) DNA extraction, 2) BestRAD analysis and sequencing, 3) Genetic diversity analyses, 4) Population differentiation estimation, 5) Movement pattern estimation, and 6) Captive breeding simulations. Quarterly reports will be provided to collaborators to document progress in achieving milestones.

We will disseminate materials and prepare a presentation for our collaborators at the Hawaiian Monk Seal Research Team. By working with the team throughout the project, we can ensure that our results maximize their impact on Hawaiian monk seal recovery. The genetic component of a captive breeding program, while important, is not the only consideration when designing a captive breeding program and thus we need to coordinate with the whole group on this effort. We plan to publish our approach for utilizing genetic data to guide the establishment of a captive breeding program in a peer-reviewed journal, and to disseminate our materials (laboratory protocols, bioinformatic pipelines, code) to others interested in using genetic data as a guide for establishing a captive breeding program. Interested groups might include those focused on utilizing molecular data for conservation - AZA's Molecular Data for Population Management Scientific Advisory Group, The Wildlife Society's Molecular Ecology Working Group, the Society for Conservation Biology's Conservation Genetics Working Group, and IUCN's Conservation Genetics Specialist Group.

8. Explain and identify the means, and timeline of how the key results and lessons of the project will be disseminated and shared. Key internal and external audiences should be identified. **LIMIT 1,500 CHARACTERS with spaces**

There are several avenues by which we will disseminate our results. Our results for the wild Hawaiian monk seal populations (Objective 1) will be shared with our collaborators (NOAA-NMFS Hawaiian Monk Seal Recovery Program), published in a peer-reviewed journal, and presented at a research conference (e.g., AZA, Society for Conservation Biology). We will also prepare an infographic and a short public interest piece that can be used by conservation organizations (e.g., Hawaii Marine Animal Response, Marine Mammal Center), news outlets, and facilities with existing Hawaiian monk seals (including the Waikiki Aquarium and the AZA-accredited Minnesota Zoo) that explains our findings and their importance for Hawaiian monk seal recovery, in terms that are accessible for the general public. We will continue to participate in ongoing discussions regarding translocations, head start programs, and other management activities to benefit Hawaiian monk seal recovery.

Our plan for a Hawaiian monk seal captive breeding program (Objective 2) will be shared with the Recovery Program, and we will coordinate with them to integrate our genetic considerations into a more comprehensive captive breeding plan. Our generalized framework for utilizing genetic data to guide captive breeding program establishment will be published in a peer-reviewed journal, and will be disseminated directly to organizations with an interest in utilizing molecular data for captive breeding and species conservation

9. Have all the necessary permits been obtained for the project?

a) ☐ Yes ☐ No ☒ Not Applicable

If yes, list the permits here and attach one copy of each to your CGF Application email. Attachments should be titled **"Permit #."**

Not applicable

- b) If the necessary permits have not yet been obtained, please explain why. **Pending permits must be obtained and submitted to AZA by 15 June 2020.**

Not applicable

10. If research animals are to be purchased for the project, indicate and justify the destination of the animal(s) after completion of the research. **LIMIT 750 CHARACTERS with spaces**

Not applicable

11. If this is a multi-year project, identify the funding strategy for subsequent years. **LIMIT 750 CHARACTERS with spaces**

Not applicable

12. **Past CGF Support.** If any of the proposal's Principal or Co-Investigators listed on the Cover Sheet have received previous CGF support, complete the required information below for each.

a) Name of Investigator: Not applicable Year of Award: Not applicable Amount of Award: Not applicable

Project Title: Not applicable

Were all required reports (annual/final) submitted? ☐ Yes ☐ No

b) Name of Investigator: Year of Award: Amount of Award:

Project Title:

Were all required reports (annual/final) submitted? ☐ Yes ☐ No

Note: If additional awards need to be included, add their information to a separate document titled “**Past CGF Projects**” and include it as an attachment with your CGF Application email.

2020 AZA CONSERVATION GRANTS FUND

Project Budget

BUDGET CYCLE

All funding requests from CGF must be for a maximum period of one year only. Funding decisions are announced in early September and expenses incurred prior to the award date will not be covered by CGF funds. CGF funds are to be used within one year from the project start date and forward.

DETAILED BUDGET

A detailed budget sheet is provided below and information should be separated into CGF and non-CGF costs. List under each category the budget items and method of calculation for each (e.g., 2 persons x 10 days x \$25/day), as well as probable sources of support. The following should be listed as **separate** budget items:

- airfares
- lodging
- vehicle rental (automobiles, planes, boats)
- fuel
- per diem
- equipment over \$500
- supplies, services (lab analyses, editing)
- pharmaceuticals
- shipping/postage
- printing
- personnel costs

CGF does not fund the following: Tuition, university fees, or fringe benefits associated with graduate students or other project personnel; administrative costs (including institutional overhead and submission costs for the publication of journal articles); travel unrelated to completion of the project (conference presentation expenses, travel to meetings, etc.); planning meetings that define, rather than implement, conservation/management/education goals.

CGF does not fund salaries for project personnel other than graduate students, technicians, and start-up positions established solely for the purpose of completion of the project.

Note: Recipients of CGF grants are obligated to inform AZA (CGF@aza.org) of any pending or overlapping grants received. If such grants are received, AZA's CGF Scientific Advisory Committee may recommend that the CGF award amount be reduced or refunded.

Note: If additional budget items need to be included, add this information in a separate document titled "**Additional Budget Items**" and include it as an attachment with your CGF Application email.

Budget Items Form:

Budget Item	Method of Calculation (e.g., 2 persons x 10 days x \$25/day)	CGF- Requested Amount	Non-CGF Requested Amount (if applicable)	Date Expected from Non-CGF Source (if applicable and not yet received)	Name of Non-CGF Source (e.g., Grant Fund, Matching)	Total Expense (CGF+Non-CGF)
DNA extraction	\$4.18 per sample X 250 samples	\$1,045	\$n/a	n/a	n/a	\$1,045
BestRAD library prep	\$50 per sample X 200 samples	\$10,000	\$n/a	n/a	n/a	\$10,000
Illumina HiSeq Reun	\$3,650 per sequencing lane X 2 lanes	\$7,300	\$n/a	n/a	n/a	\$7,300
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	TOTAL PROJECT BUDGET (round to nearest whole dollar)	\$18,345	\$			\$18,345

BUDGET JUSTIFICATION

State why each item listed in the CGF Budget Items Form costs what it does and indicate how each item relates to the proposed scope of work. Sufficient detail should be provided to address any potential reviewer concerns with respect to cost and need. If the lifespan of equipment costing over \$500 exceeds that of the grant period, detail and justify who will assume ownership of the item. **LIMIT 2,500 CHARACTERS with spaces**

We are requesting funds to perform genetic analysis of 200 Hawaiian monk seal samples, to meet the objectives outlined in our proposal. All budgeted items fall in the category of 'Supplies, Services'; this includes DNA extraction, BestRAD library preparation, and next-generation sequencing (details below). Existing funds are covering the cost of the complementary project, and we will use the SNP panel and a subset of the data produced by that project in this project. Funds are not being requested for computing resources to undertake the bioinformatics or simulation portions of the project. The University of Wisconsin-Milwaukee has a high-performance research computing cluster that we can use for this purpose.

Supplies, Services:

1. DNA extraction: per sample price for Qiagen's DNeasy Blood and Tissue Kit was calculated from Qiagen's website (\$2.88 per sample) plus additional consumables required for the protocol (tips, tubes, plates, total \$0.50 per sample). Qubit quantification requires an assay kit and tubes (\$0.80 per sample). Total cost of DNA extraction supplies is \$4.18 per sample, or \$1,045 for the project (200 samples plus 50 potential re-extractions).
2. BestRAD library preparation: per sample price was tabulated from necessary supplies as per published protocol in Ali et al (2016), including barcodes, adapters, restriction enzymes, and reagents for the ligation, PCR, and cleanup steps (\$50 per sample). Total cost of BestRAD library preparation supplies is \$10,000 for the project.
3. Illumina Hi-seq run: price of a single paired-end sequencing lane as per University of Wisconsin-Madison's Biotechnology Center (as internal users we pay reduced cost; \$3,650 x 2 lanes). Total cost to the project is \$7,300.

2020 AZA CONSERVATION GRANTS FUND

Proposal Resubmission Form

If a proposal for this project was previously submitted for Conservation Grants Fund (CGF) funding and rejected, this form must be completed and attached to the CGF Application email, titled “**Resubmission Form**” (see CGF Application Instructions).

Previous CGF Proposal Title: Improved Hawaiian monk seal conservation through genomics

Year Submitted: 2019

State how this version of the proposal addresses the previous reviewers' comments. Please be thorough in your explanation. **LIMIT 12,000 CHARACTERS with spaces**

Reviewer 1

This was a thoughtfully written and clear proposal. I appreciate that the main goals were reiterated in various sections throughout the document. I needed to brush up on my genetics by looking up SNP on the internet so maybe other reviewers would have liked a brief definition too.

A huge strength of this proposal is that the tissue samples required have already been collected so that removes one major obstacle towards the successful completion of the project. Using tissue samples for as many research projects as possible is a thoughtful use of resources. The staff wages and equipment necessary for the study are already funded through other complimentary studies so the request for the funding of supplies and lab tests to support this initiative is another notable efficiency.

I appreciate that you will make you resources – laboratory protocols, bioinformatic pipelines, codes developed – available to the AZA community since this could be important as we move forward with captive population sustainability initiatives with other phocid species.

A very small weakness of the proposal is that the emphasis of the use of genetic data to guide the establishment of a captive breeding program seems a bit narrow in the absence of at least mentioning that its only one component of a multipronged approach including behavioral and husbandry expertise.

It refreshing that NOAA and the PIs are being proactive about considering a breeding program before the population gets too low (a.k.a. Vaquita). They are also a head of the game about understanding the genetic profile for both in situ and ex situ management. It is a straight forward proposal with important implications. I would suggest to the investigators consider their audience when writing grants. This is full of genetic jargon which would benefit from some explanation around the terminology for reviewers that do not have as strong as background in the subject (i.e. they never explain what an SNP is other that it is a variant). It may seem trivial but when competing against so many proposals these little details count.

Response to Reviewer 1:

Thank you for your comments and taking the time to review our grant proposal.

We appreciate the suggestion regarding genetic jargon and terms. We strived to improve readability by both reducing genetic jargon and providing more explanation throughout. In an effort to maintain transparency in our methods, we retained details such as software programs or statistical approaches in the Methods section only. We agree that genetic data is but one component of a successful captive breeding program, and have revised our proposal to emphasize the importance of combining the data we will generate on optimal number and representation of founders with husbandry and behavioral information to actually establish a program.

Reviewer 2

The proposal was well-written, succinct, and a well-argued approach to use genomics to inform seal recovery, written by an extremely well qualified team.

The project sufficiently connects with a critical conservation need (characterizing genetic variation that can be used for in situ management decision-making) for an important species, and would be a unique opportunity to have an evidence-based founding of a captive breeding program for the species. Of a bit of concern, however, was that I could not find evidence of captive breeding being proposed as a recovery strategy in the 2007 Recovery Plan or the 2016 Hawaiian islands management plan. If the PIs had provided more evidence of definitive interest from NOAA and a timeline for when the captive program was being considered would be more supportive for that part of the project's goals.

The PIs could have done a better job clarify what current management strategies might be informed by their results – are there currently translocations between islands, on what frequency and based on what criteria? If a population was found to have low genetic diversity or high inbreeding, what actions would NOAA take? That link could have been made more directly.

In the methods, the molecular methods are clearly described. The designing of breeding simulations to assess optimal number and representation of founders and simulations for diversity retention is described very vaguely – will this modeling be done using software like PMx, Vortex, or custom-build models? The methods seem slim in this area.

I liked the PI's plan to disseminate materials as a guide to other programs that may be interested in using a genomic approach when forming a captive breeding program for conservation.

Response to Reviewer 2:

We appreciate your comments on our grant proposal, and thank you for your review.

There is no formal documentation for NOAA's interest in a captive breeding program for Hawaiian monk seals. However, the need for a captive breeding program, and the current status of our understanding of husbandry, reproduction, and behavior, have been discussed during a series of meetings over the last 2 years. These meetings were led by our collaborating institute, NOAA's Hawaiian Monk Seal research program, and included Hawaiian monk seal researchers and managers from AZA facilities housing Hawaiian monk seals (e.g., Minnesota Zoo). In our revised proposal, we have outlined NOAA's interest in a captive breeding program and provided an Appendix from the Minnesota Zoo to illustrate their support for foundational research for a future captive breeding program.

We have also added more information regarding the current management strategies for the Hawaiian monk seal and how our research could improve conservation efforts.

We have also clarified the breeding simulations and provided the software that we plan to use (PMx).

Reviewer 3

This project address both in situ and ex situ conservation needs of the critically endangered Hawaiian monk seal in a collaborative effort between AZA researchers and NOAA-NMFS. The proposal makes a clear case for the value of this work and the urgency for completing it. The descriptions of laboratory and data analysis methodologies are clear and the plan seems likely to provide informative and valuable results. The only suggestion I would make would be to allow more time for data processing and analysis, and to include a buffer in the budget in case samples need to be re-run.

Response to Reviewer 3:

Thank you for taking the time to provide helpful feedback on our grant proposal.

We have expanded the time allotted for data processing and analysis, while still aiming to complete the project within the allowed 1-year grant period. This revised schedule is in line with other ongoing projects in our laboratory that are generating and analyzing similar types of data. We have also included additional buffer in the budget to allow for sample re-runs; our previous work for a variety of other species suggests that we might expect ~5% of samples to be re-run.

Reviewer 4

This is a good proposal that has specific results within its first year, focuses on a highly endangered species, and provides needed information to make informed conservation decisions on managing the wild population.

Response to Reviewer 4:

We appreciate your comments and the time that you spent to review our grant proposal.

Reviewer 5

Only negative is that seals are not a part of SAFE. Emphasis on genetics for breeding program seems a bit narrow.

Response to Reviewer 5:

Thank you for your comments and taking the time to review our grant proposal.

We agree that genetic data is but one component of a successful captive breeding program. We have revised our proposal to emphasize the importance of combining the data we will generate on optimal number and representation of founders with husbandry and behavioral information to actually establish a program.

Citations

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Samantha S. Hauser

Postdoctoral Fellow, Department of Biological Sciences
University of Wisconsin - Milwaukee

EDUCATION

PhD (2018)	University of Louisiana at Lafayette, <i>Environmental and Evolutionary Biology</i>
BS (2013)	Rutgers University, <i>Ecology and Natural Resources & Environmental Geomatics</i>

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow (2018 – Present)	University of Wisconsin- Milwaukee
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PUBLICATIONS

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4. Seaborn, T., **Hauser S.**, Konrade, L., Waits, L., & Goldberg, C. **(2019)** Landscape-level sampling and inference from individuals versus populations. *Ecology & Evolution*
 3. **Hauser S.**, Walker L., Leberg, L. **(2019)** Asymmetrical gene flow of the recently delisted passerine black-capped vireo (*Vireo atricapilla*) indicates source-sink dynamics in central Texas. *Ecology & Evolution*
 2. **Hauser, S.**, Wakeland, K., and Leberg, L. **(2019)** Inconsistent use of multiple comparison corrections in studies of genetic spatial structure: Are some Type I errors more tolerable than others? *Molecular Ecology Resources*
 1. **Hauser, S.**, Meixler, M. S., & Laba, M. **(2015)** Quantification of Impacts and Ecosystem Services Loss in New Jersey Coastal Wetlands Due to Hurricane Sandy Storm Surge. *Wetlands*, 35(6), 1137-1148.

GRANTS/AWARDS

Fellowships/Assistantships (\$96,000+):

University of Louisiana at Lafayette Teaching Assistantship (\$7,500/semester)	2017- 2018
Department of Defense Research Assistantship (\$3,000/summer)	2014-2016
University of Louisiana at Lafayette Fellowship (\$18,000/year)	2013-2017

Small Grants (\$1,700+):

University of Louisiana Graduate Student Organization Research Grants	2013-2016, 2018
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Awards (\$4,200+):

AOS 2018 Conference Travel Award (\$382.50)	2018
University of Louisiana Graduate Student Organization Travel Awards (\$300 each)	2017-2018
University of Louisiana Graduate Student Organization Travel Awards (\$500 each)	2016-2018
University of Louisiana Student Government Association Travel Award (\$200)	2017-2018
Diversity in Ornithology Honorable Mentions, AOS/SCO 2017 Conference (\$225)	2017
AOS/SCO 2017 Conference Travel Award (\$510)	2017
United States International Association of Landscape Ecologist (US-IALE) Honorable Mention for Best Student Presentation (\$150)	2017
United States International Association of Landscape Ecologist (US-IALE) Student Travel Award (\$500)	2017
H. Dickson Hoese & Richard Moore Award for Best Graduate Student Publication Award (\$150)	2016

PRESENTATIONS

I have given 10 presentations of my research at professional meetings, and have been a co-author on an additional 2 talks. I also have given 1 invited seminar at a university.

PROFESSIONAL SERVICE

Molecular Ecology Resources Reviewer (2018- Present)

Molecular Ecology Reviewer (2018- Present)

Landscape Ecology Reviewer (2019- Present)

Conservation Genetics Reviewer (2019-Present)

British Ornithological Union Twitter Conference Coordinator (2018)

American Ornithological Society Conference Molecular Ecology Symposium Chair (2018)

British Ornithological Union Twitter Conference Symposium Chair (2018)

Joint meeting of American Ornithological Society and Canadian Ornithological Society

Conference Conservation Genetics Symposium Chair (2017)

17th Annual Graduate Student Symposium Committee Chair (2018)

16th Annual Graduate Student Symposium Committee Chair (2017)

MENTORING EXPERIENCE

I have mentored 3 undergraduate researchers during my PhD, first through molecular techniques and then through their own independent research projects.

TEACHING EXPERIENCE

I have taught courses as a teaching assistant ranging from a 100-level undergraduate courses to 500-level graduate courses, including Fundamentals of Biology Laboratory (100-level), Biology for Elementary/Middle Education Majors (200-level), Ornithology (400/500 level), and Spatial Data and Management (500-level). In addition to my teaching assistant role, I was made an adjunct instructor during my graduate career to teach the Fundamentals of Biology (100-level course) laboratory course autonomously.

PUBLIC OUTREACH

I have been involved in public outreach events since 2013, namely through the Louisiana Audubon Bird Observatory (2013-2018) and Annual Audubon Yellow Rails and Rice Festival (2014-2016). I also contributed to a K-12 STEM camp in Lafayette, LA through Project Front Yard. I curate and run the science communication twitter game “#BirdQuiz” that teaches bird identification weekly (2017-Present). I co-founded and co-moderate a diversity in STEM campaign called ‘46 Questions for Scientists’, for which we interview and promote scientists, particularly under-represented minorities, to make scientists more relatable to the general public and showcase diversity in STEM. I also participate in Skype a Scientist (7 talks) in which I gave a presentation about my work to middle-high school classrooms and answered questions about conservation genetics, Hawaiian monk seals, and science.

Emily K. Latch

Associate Professor, Department of Biological Sciences
University of Wisconsin-Milwaukee

DEGREES AND PROFESSIONAL EXPERIENCE

2014 – present	Associate Professor, University of Wisconsin-Milwaukee
2008 – 2014	Assistant Professor, University of Wisconsin-Milwaukee
2007 – 2008	Postdoctoral Fellow, Smithsonian Institution's National Zoological Park
2006 – 2007	Visiting Assistant Professor, Eastern Illinois University
2004 – 2006	Postdoctoral Fellow, Purdue University
1999 – 2004	PhD, Population Genetics, Purdue University
1995 – 1999	BS, Genetic Engineering, Cedar Crest College

PUBLICATIONS (last 5 years; for full publication list see my [Google Scholar page](#))

- 1) Haines ML, Luikart G, Amish SJ, Smith S, Latch EK (2019) Evidence for adaptive introgression of exons across a hybrid swarm in deer. *BMC Evolutionary Biology*, 19, 199.
- 2) Martchenko D, Prewer E, Latch EK, Kyle CJ, Shafer ABA (2019) Population genomics of ungulates. In: *Population Genomics: Wildlife* (ed Hohenlohe P), Springer Publishing, New York. *In press*.
- 3) Oyler-McCance SJ, Latch EK, Leberg PL (2019) Conservation genetics and molecular ecology in wildlife management. Chapter XX in *The Wildlife Techniques Manual* (ed: Silvy NJ). *In press*.
- 4) Jost L, Archer F, Flanagan S, Gaggiotti O, Hoban S, Latch EK (2018) Differentiation measures for conservation genetics. *Evolutionary Applications*, 11, 1139-1148. ***6th most downloaded article in 2018**
- 5) Flanagan S, Forester B, Latch EK, Aitken S, Hoban S (2018) Guidelines for planning genomic assessment and monitoring of locally adaptive variation to inform species conservation. *Evol Appl*, 11, 1035-1052. (Special Issue: Next-gen conservation genetics). ***3rd most downloaded article in 2018**
- 6) Giglio RM, Ivy JA, Jones LC, Latch EK (2018) Pedigree-based genetic management improves bison conservation. *Journal of Wildlife Management*, 82, 766-774.
- 7) Hapeman PA, Latch EK, Rhodes OE, Swanson BJ, Kilpatrick CW (2017) Genetic Population Structure of Fishers in the Great Lakes Region: Remnants and Reintroductions. *Canadian Journal of Zoology*, 95, 869-876.
- 8) Mulder KP, Walde AP, Boarman WI, Woodman AP, Latch EK, Fleischer RC (2017) No paternal genetic integration in desert tortoises following translocation. *Biol Cons*, 210, 318-324. ***2018 Stebbins Award winner**
- 9) Kierepka EM and Latch EK (2016) High gene flow in the American badger overrides habitat preferences and limits broadscale genetic structure. *Molecular Ecology*, 25, 6055-6076.
- 10) Powell JH, Cosart T, Amish SJ, Haynes GD, Luikart G, Latch EK (2016) Identifying SNPs and candidate genes associated with lineage divergence: use of next-generation targeted resequencing in mule deer. *Molecular Ecology Resources*, 16, 1165-1172 (Special Issue: Sequence Capture)
- 11) Martinsen E, McInerney N, Brightman H, Ferebee K, Walsh T, McShea W, Forrester T, Ware L, Joyner P, Perkins S, Latch EK, Yabsley M, Schall J, Fleischer R (2016) Hidden in plain sight: Endemic malaria parasites in the iconic North American white-tailed deer. *Science Advances*, 2, e1501486. ***press incl. Newsweek, Science**
- 12) Giglio RM, Ivy JA, Jones LC, Latch EK (2016) Evaluation of alternative management strategies for maintenance of genetic variation in wildlife populations. *Animal Conservation*, 19, 380-390.
- 13) Kierepka EM, Latch EK (2016) Fine-scale landscape genetics of the American badger (*Taxidea taxus*): disentangling landscape effects and sampling artifacts in a poorly understood species. *Heredity*, 116, 33-43.
- 14) Kierepka EM, Latch EK (2015) Performance of partial statistics in individual-based landscape genetics. *Molecular Ecology Resources*, 15, 512-525.
- 15) Anderson S, Kierepka E, Swihart R, Latch EK, Rhodes O (2015) Assessing permeability of landscape features to animal movement: Using genetic structure to infer functional connectivity. *PLoS ONE*, 10(2): e0117500.
- 16) Mowry RA, Schneider TM, Latch EK, Gompper ME, Beringer J, Eggert LS (2015) Genetics and the successful reintroduction of the Missouri river otter. *Animal Conservation*, 18, 196-206.
- 17) Latch EK, Reding DM, Heffelfinger JR, Alcalá-Galvan CH, Rhodes OE (2014) Range-wide analysis of genetic structure in a widespread, highly mobile species (*Odocoileus hemionus*) reveals the importance of historical biogeography. *Molecular Ecology*, 23, 3171-3190.
- 18) Hapeman P, Latch EK, Rhodes OE, Kilpatrick CW (2014) When recent and evolutionary histories meet: deciphering temporal events from contemporary patterns of mtDNA from fishers (*Martes pennanti*) in north-eastern North America. *Journal of Zoological Systematics and Evolutionary Research*, 52, 331-337.

GRANTS (active during last 5 years included)

- 2018-2021 *Genomic analysis of population history and reproduction for Hawaiian monk seals.*
NOAA/National Marine Fisheries Service.
- 2017-2019 *Landscape genomics for conservation of the Cascades frog.*
UWM Research and Creative Activities Support.
- 2016-2019 *Snowshoe hare density in Michigan's UP estimated using genetic mark-recapture of noninvasively collected fecal samples.*
Sault Ste. Marie Tribe of Chippewa Indians.
- 2015-2020 *Developing new tools for integrating DNA data into AZA breeding programs.*
Institute of Museum and Library Services (IMLS) – National Leadership Grant.
- 2015-2018 *Unraveling plague ecology through pathogen, vector, and host genetics.*
Morris Animal Foundation.
- 2015-2020 *Evaluating regional and landscape-scale movement patterns of animal populations using genetic approaches.*
US Forest Service.
- 2015-2018 *A landscape genomics framework to study introgression in non - model species.*
UWM Research Growth Initiative.
- 2013-2015 *Pilot project estimating association of size-structure decline of bluegill with population genetic changes.*
Wisconsin Department of Natural Resources.
- 2013-2016 *Reproduction in lake trout morphotypes surrounding Isle Royale.*
Great Lakes Fishery Commission.
- 2010-2016 *Evaluation of alternative culling strategies on maintenance of genetic variation in bison.*
US Fish and Wildlife Service.
- 2010-2013 *Speciation caught in the act: Identification of genes under selection in a vertebrate hybrid zone.*
UWM Research Growth Initiative.
- 2010-2011 *Using molecular gene diversity estimates to improve genetic management and reintroduction potential of the Addax SSP.*
San Diego Zoo Global.
- 2009-2015 *Assessment of the status of American badger in Wisconsin using genetic and ecological techniques in a GIS framework.*
Wisconsin Department of Natural Resources.

PRESENTATIONS AND INVITED SEMINARS

During the past five years, I have given 18 presentations of my research at professional meetings, and have been a co-author on 35 talks and posters. I was an invited symposium speaker at 5 of these conferences. I also have given 10 invited seminars at universities and organizations over the past 5 years.

ADVISING

I have served as major advisor for 4 postdoctoral fellows, 4 PhD students, 5 MS students, and 10 undergraduate students since my arrival at UWM in 2008. I have served on 18 graduate student committees during that time period.

TEACHING EXPERIENCE

In my current position at the University of Wisconsin-Milwaukee, I teach an average of over 500 student credit hours per year, in large-enrollment undergraduate classes, upper-level undergraduate/graduate courses, and advanced graduate courses. All have reflected the highest quality of instruction, with average evaluation scores of 4.69, 4.88, and 4.94 out of 5.00, respectively.

SELECTED ACTIVITIES

- Invited member, IUCN Conservation Genetics Specialist Group (2016-present)
- Invited member, Assoc. of Zoos & Aquariums (AZA) Scientific Advisory Group – Molecular Data for Population Management (2017-present)
- Invited speaker, National Academy of Sciences Committee on Wolf Taxonomy (2018)
- Editorial Review Board, Molecular Ecology (2012 – present), named “Top Reviewer” in 2014-2018
- Molecular Ecology Working Group, The Wildlife Society (co-founder, 2014; Pres-Elect/President, 2014-2019)
- Invited participant, NIMBioS Investigative Workshop: Next Generation Genetic Monitoring (2016)

2020 AZA CONSERVATION GRANTS FUND

Statement of Institutional Support

One copy of this statement must be **electronically signed** by the institutional CEO/Director using the checkbox below, and submitted as an attachment to the CGF Application email titled “**SIS Institution Name**” for *each* institution or organization that fits one or more of the following criteria:

1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

By signing this form, all participating institutions, whether participating as a facility or through an employee, take responsibility for: 1) completing their part of the proposed project and 2) obtaining Institutional Animal Care and Use Committee (IACUC)* or Institutional Review Board (IRB)** approvals, if necessary. All accountability for conducting responsible research lies in the hands of each participating institution, not the CGF Scientific Advisory Committee.

All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) NOAA Hawaiian Monk Seal Research Program has reviewed the proposal titled (*Proposal Title*) “Improved monk seal conservation through genomics” and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA’s Conservation Grants Fund.

Michelle Barbieri Lino

Name

Program Lead, NOAA Hawaiian Monk Seal Research Program

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

3/12/2020

Date

X Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.

* Institutional Animal Care and Use Committee (IACUC, for U.S. applicants) or the equivalent oversight committee (for non-U.S. applicants) approval is desirable for any proposal, but is required for any project in which animals are to be involved in invasive procedures and/or procedures that may result in pain, harm, distress, and/or altering of behavior (this includes any time an animal in the wild is sedated or anesthetized). This generally excludes samples obtained during regularly scheduled veterinary or husbandry procedures or emergency medical procedures. IACUC (or the international equivalent thereof) approval is required for both *in situ* and *ex situ* projects. Each institution involved in a project must submit the project proposal to its IACUC (or equivalent) for review. In order to be considered complete, proposals must include the Statement of Institutional Support from each participating institution/organization as recognition of compliance with these IACUC (or equivalent) approval guidelines.

If an institution is not registered as a research facility or similar facility and therefore does not have an IACUC (or equivalent), it needs to become affiliated with a university, zoo, or other body that has an IACUC (or equivalent) in order to obtain approval.

** Any projects supported through CGF funding that involve human subjects must be conducted under the auspices of a federally registered Institutional Review Board (IRB) and be consistent with 45 CFR Part 690: Federal Policy for the Protection of Human Subjects if the work or researchers are based in the USA. For members submitting from outside the USA for projects to be completed entirely outside the USA and its territories may use a process that meets the requirements of the prevailing legislation for protection of human subjects where the principal investigator is based or in the country where data collection will occur, whichever is more stringent. We note that all protocols must still conform to the types of protections outlined in 45 CFR Part 690 irrespective of the reviewing organization.

2020 AZA CONSERVATION GRANTS FUND

Statement of Institutional Support

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1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
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All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) University of Wisconsin-Milwaukee has reviewed the proposal titled (*Proposal Title*) Improved Hawaiian monk seal conservation through genomics and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Kate Mollen

Name

Interim Director, Office of Sponsored Programs, University of Wisconsin-Milwaukee

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

February 25, 2020

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

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